

Alice Wang

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EDUCATION

University of Chicago, Chicago, IL, US

September 2023 — June 2027

B.S. in Physics, B.S. in Mathematics

Cumulative GPA: 3.75/4.00

Relevant Coursework: Advanced Classical Mechanics (PHYS 316)*, Advanced Electrodynamics and Optics I-II (PHYS 322-323)*, Statistical Mechanics (PHYS 352)*, Quantum Field Theory I-II (PHYS 443-444)*, Introduction to Representation Theory of Finite Groups (MATH 267)

** denotes graduate course*

Campus involvements: UCQuantum Head of Education, UC Geography Events Secretary, University Symphony Orchestra Violist

EXPERIENCE

Pritzker School of Molecular Engineering

Chicago, IL, US

Undergraduate Researcher in Maurer Lab

December 2023 — Present

- Programmed instrumentation control sequences for wide-field nuclear magnetic resonance (NMR) imaging.
- Developed data analysis pipeline for live tracking of nanodiamond (ND) trajectories embedded in moving cells.
- Designed and performed experiments for testing resolution and accuracy of confocal imaging with particle tracking techniques.
- Developed computer simulations and mathematical models for imaging "barcoded" cells containing nitrogen vacancy (NV) centers.

MemQ

Chicago, IL, US

Jeff Metcalf Intern

June 2025 — September 2025

- Conducted extensive literature review of periodically-poled lithium niobate (PPLN) waveguides for quantum memory applications.
- Implemented periodic-poling in lithium niobate (LN) waveguides by programming signals in a high-voltage amplifier.
- Programmed automated algorithm for optically probing second harmonic generation (SHG) by waveguides in photonic chips.
- Designed photonic chips and micro-scale waveguide elements for mass foundry fabrication using kLayout.

University of Chicago Math REU

Chicago, IL, US

Full Program Participant

June 2025 — September 2025

- Wrote and presented original expository paper on quantum error correction with the toric code (2024- Apprentice Program).
- Wrote original expository paper on stochastic differential equations modeled in first-passage percolation (2025- Full Program).
- Attended lectures and problem sessions on various mathematical topics, especially for probability theory and algebraic topology.

Argonne National Laboratory

Lemont, IL, US

Seasonal Intern with Joosten Group

May 2024 — September 2024

- Tested and designed components of the Barrel Imaging Calorimeter for the Electron-Proton/Ion Collider (ePIC) collaboration.
- Used machine learning (Keras/Pytorch) to analyze simulated particle showers in the implementation of silicon AstroPix detectors.
- Collaborated with staff scientists and postdocs in building cosmic ray detectors for device engineering tests.
- Presented findings to colleagues in group meetings and poster project as part of Argonne's "Learning on the Lawn" event.

PUBLICATIONS

Design of Hex Beamline at the NSLS-II, Brookhaven National Laboratory

Published

- Zhong, Z., Broadbent, A., Drakopoulos, M., Dvorak, J., Hasnah, M., Holmes, D., Lucas, M., O'Hara, S., Softye, E., & Wang, A. (2021). Optical design of DEI/ABI system at the Hex Beamline at the NSLS-II. *Synchrotron Radiation News*, 34(6), 32–40. <https://doi.org/10.1080/08940886.2021.2022401>.

AWARDS

YQuantum Grand Prize Winner

Yale Quantum Institute

Placed 2nd overall at the YQuantum Hackathon (Alice and Bob Challenge), winning a cash prize of \$2500.

March 2025

AIME Qualifier

Mathematical Association of America

Qualified for the American Invitational Mathematics Examination and scored 7/15 on the exam.

February 2022

SKILLS

- **Programming:** Python (Keras/Pytorch, SciPy, NumPy, Pandas, Matplotlib), Java, R, C#, MATLAB
- **Tools:** Fusion360, Mathematica, Gwyddion, kLayout, Linux, VSCode, L^AT_EX, Jupyter
- **Languages:** English, Mandarin Chinese, Italian